

Multi: Merging WordNet/TUFS and Mapping Examples

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What we are trying to do — part 1

- **Goal** — produce a unified, multilingual merged WordNet with a reduced synset graph.
- **Why** — remove redundancy, unify identifiers, and enable automated processing of examples across languages.
- **Primary output** — unifiedSenseId for examples and structured Example elements with TokenMatch entries.

What we are trying to do — part 2 & What is WordNet

- **Multilingual coverage** — support many languages (special attention to Japanese) using spaCy and language-specific tools.
- **Lexicalization** — detect and mark multiword and lexicalized expressions in examples.
- **What is WordNet** — a lexical database grouping words into *synsets* (sets of synonyms) connected by relations (hypernymy, hyponymy, meronymy).

WordNet visualizations

```
<lexicalResource url="https://github.com/tufts/wordnet-xml" language="es" email="bord@ieee.org"
  license="https://creativecommons.org/licenses/by/4.0/" version="2.0" url="https://github.com/owm/tufts/" confidenceScore="0.95"/>
<lexicon id="tufts-es" label="TUFS Basic Vocabulary for Spanish (Español)" language="es" email="bord@ieee.org"
  license="https://creativecommons.org/licenses/by/4.0/" version="2.0" url="https://github.com/owm/tufts/" confidenceScore="0.95"/>
<lexicalEntry id="tufts-es-enaro-n">
  <lemma writerForm="enaro" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/es/vmod/sound/word/word_1069.mp3"/>
    <lemma>
      <sense id="tufts-es-enaro-15210045-n" synset="tufts-es-15210045-n">
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      </sense>
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  </lexicalEntry>
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  <lemma writerForm="estudio" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/es/vmod/sound/word/word_828.mp3"/>
    <lemma>
      <sense id="tufts-es-estudio-65755883-n" synset="tufts-es-65755883-n">
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  <lemma writerForm="marrón" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/es/vmod/sound/word/word_1189.mp3"/>
    <lemma>
      <sense id="tufts-es-marrón-64971928-n" synset="tufts-es-64971928-n">
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  </lexicalEntry>
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  <lemma writerForm="élève" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/es/vmod/sound/word/word_1143.mp3"/>
```

Figure 1: Example: spanish wordnet

```
<lexicalResource url="https://github.com/tufts/wordnet-xml" language="en" email="bord@ieee.org"
  license="https://creativecommons.org/licenses/by/4.0/" version="2.0" url="https://github.com/owm/tufts/" confidenceScore="0.95"/>
<lexicon id="tufts-en" label="TUFS Basic Vocabulary for English (English)" language="en" email="bord@ieee.org"
  license="https://creativecommons.org/licenses/by/4.0/" version="2.0" url="https://github.com/owm/tufts/" confidenceScore="0.95"/>
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  <lemma writerForm="January" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/en/vmod/sound/word/word_1118.mp3"/>
    <lemma>
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        <Count>1</Count>
      </sense>
    </lemma>
  </lexicalEntry>
<lexicalEntry id="tufts-en-study-n">
  <lemma writerForm="study" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/en/vmod/sound/word/word_809.mp3"/>
    <lemma>
      <sense id="tufts-en-study-65755883-n" synset="tufts-en-65755883-n">
        <Count>1</Count>
      </sense>
    </lemma>
  </lexicalEntry>
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  <lemma writerForm="brown" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/en/vmod/sound/word/word_1236.mp3"/>
    <lemma>
      <sense id="tufts-en-brown-64971928-n" synset="tufts-en-64971928-n">
        <Count>1</Count>
      </sense>
    </lemma>
  </lexicalEntry>
<lexicalEntry id="tufts-en-Sunday-n">
  <lemma writerForm="Sunday" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/en/vmod/sound/word/word_1131.mp3"/>
    <lemma>
      <sense id="tufts-en-Sunday-15163797-n" synset="tufts-en-15163797-n">
        <Count>1</Count>
      </sense>
    </lemma>
  </lexicalEntry>
<lexicalEntry id="tufts-en-snow-n">
  <lemma writerForm="snow" partOfSpeech="n">
    <pronunciation audio="https://www.coolang.tufts.ac.jp/mt/en/vmod/sound/word/word_1193.mp3"/>
```

Figure 2: Example: english wordnet

How we are doing it (pipeline)

- **Pass 2** — parse XML, extract Lemma and build `sense_to_lemma_map`.
- **Pass 3** — merge and reduce synsets: normalize lemmas, deduplicate senses, assign unified IDs, apply reduction rules.
- **Pass 4** — NLP matching: normalize text, use `PhraseMatcher` for multiword expressions, create `Example` and `TokenMatch` (lemma, offsets, POS, lexicalized flag).
- **Tech stack** — Python, `spaCy`, `sudachipy` (Japanese), `lxml/etree`.

Code versions and differences

- **Version A (older)** — token-level matching using `token.lemma_`; lazy spaCy loading; builds `sense_to_lemma_map`; no merge/reduction step; no multiword matcher; no explicit Unicode normalization.
- **Version B (upgraded)** — adds Unicode normalization, `PhraseMatcher` for multiword lemmas, explicit `pass3_merge_and_reduce` producing unified IDs, `sudachipy` support for Japanese, and a `lexicalized` flag on `TokenMatch`.
- **Practical impact** — Version B increases recall for multiword and non-ASCII lemmas and provides `unifiedSenseld` for downstream linking; Version A is simpler and faster for small datasets but misses multiword/normalized matches.

Results and problems

- **Achieved** — extraction of sense→lemma map; generation of Example elements with token offsets, POS, and lexicalized flag (in upgraded code).
- **Metrics** — internal log tracks `total_sentences`, `matches_found`, `matches_failed`, and per-language breakdown.
- **Problems** — availability and accuracy of language models; lemma mismatches due to inflection/transliteration; false negatives from tokenization; need for manual QA on failed cases.

Next steps (action plan)

- Generate a per-language report from the mapper log and identify top languages with high failure rates.
- Implement and test `pass3_merge_and_reduce` with concrete reduction rules and unified ID assignment; run end-to-end on a small sample.
- Export a sample of failed examples (priority: Japanese) for manual correction and to refine normalization and model selection.

Acknowledgements & References

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- **References** — GWC 2014 proceedings (paper on WordNet merging); CygNet project (Rowan H. M.); spaCy documentation; SudachiPy documentation.

Thank you for your attention.